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ABSTRACT

Introduction Side-to-side strength differences are often used in return to sport (RTS) decision making following hamstring injury. In football (soccer), there is little consensus, and a side-to-side difference of <10% is often suggested as a criterion. To date, no study has determined whether differences exist when comparing the strength of the injured limb at RTS to the pre-injury level. **Purpose** The aim of this study was to compare the isokinetic strength at RTS to pre-injury strength in the injured limb, and to investigate the side-to-side differences at RTS. **Methods** This study was based on a pooled group of patients participating in two single-site randomized controlled trials. All eligible professional male soccer players who suffered an MRI-positive hamstring injury were included. The isokinetic strength of the injured limb at RTS was compared to pre-injury screening tests, and the side-to-side difference was determined at both time points. **Results** In total, 41 hamstring injuries were included in the analyses. Side-to-side differences in isokinetic strength at RTS were similar to pre-injury levels. At RTS, the average strength of the injured limb was >95% compared to pre-injury strength. Overall, 63.4% of the players had a >10% negative strength difference (i.e. the injured limb being weaker) and 57.9% of the players had a >10% positive strength difference (i.e. the injured limb being stronger) in at least one of the five isokinetic variables. **Conclusion** The distribution of isokinetic strength differences when comparing strength at RTS to pre-injury measures were similar, urging clinical caution when interpreting these results. Small changes in isokinetic strength of the injured limb at RTS compared to pre-injury strength suggest that isokinetic cut-off values are not useful to guide restoration of strength as a criterion for RTS after hamstring injury. **Keywords** hamstring injuries; return to sport; criteria; isokinetic; rehabilitation

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INTRODUCTION

Hamstring injuries are common in sports that involve high speed running such as football (soccer), American football, Australian football, rugby and sprinting.(1,2) In elite soccer, muscle injuries represent 18% to 37% of all time-loss injuries,(3–5) and hamstring muscle injuries have become the most common injury reported.(4) When faced with return to sport (RTS) decisions following hamstring injury, clinicians, technical staff, and players engage in a type of risk-benefit analysis.(6) The StARRT framework(7), adapted from the original Creighton model,(8) suggests the assessment of health and activity risk, and then an assessment of risk tolerance, to determine RTS readiness. A recent systematic review of RTS after hamstring injuries demonstrated that only half of the studies included provide some definition of readiness for RTS, with a wide variety of non-validated criteria used to support the decision.(9) Muscle strength, in particular eccentric hamstring strength, is considered an important criterion when determining whether a player is ready to RTS after hamstring injury.(10–12) However, in a worldwide Delphi study conducted to address discrepancies related to RTS criteria after hamstring injuries in soccer,(13) no consensus was reached to include "similar eccentric strength" as a criterion, with the expert panel divided. The value of muscle strength tests is unclear.

In elite sports, isokinetic dynamometry tests are widely used as criteria for RTS and during athlete screening (periodic or pre-participation health evaluations), where strength is typically compared between the injured and uninjured limb. Isokinetic strength is one of the top three screening tests used at elite level in soccer,(14) even though evidence for this practice is lacking.(15) Although typically not available in general practice, the addition of isokinetic testing as a RTS criterion has been suggested,(16) however, it remains unclear if this would actually lead to more timely RTS and lower re-injury rates. Furthermore, it is often

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recommended that the strength of the injured muscle is comparable to pre-injury levels, or at least approximating the strength of the uninjured limb.(7) Although no uniform recommendation exists, 10% is often considered a benchmark for the injured muscle when compared to the contralateral uninjured side.(10,11,13)

A previous study found isokinetic testing as a RTS criterion after hamstring injury suspect, reporting that 67% of the players with a side-to-side deficit of >10% in at least one isokinetic measure at RTS.(17) The main limitation of this and other similar studies is the assumption that the uninjured side represents the pre-injury ("healthy") strength status of the injured leg. As hamstring strength might be considered a risk factor for hamstring injury,(15,18) it is possible that the observed side-to-side difference was already present preceding the hamstring injury, or perhaps the uninjured limb was strengthened during rehabilitation. However, there are no studies comparing results from pre-injury screening data to strength at RTS. Also, the minimal detectable change (the smallest amount of change required to distinguish a true performance difference from variability due to measurement error) in isokinetic tests has been reported as high as 20.6% and 24.0% for quadriceps and hamstrings muscles, respectively.(19) Therefore, the ability to interpret a difference in hamstring strength is also limited by test accuracy.

Thus, the purpose of this study was to determine whether professional soccer players who had suffered MRI-positive acute hamstring injuries displayed the same level of strength as measured during pre-season baseline testing. We aimed to compare isokinetic strength of the injured limb at RTS to pre-injury strength measures, and to assess the side-to-side differences at RTS.

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METHODS

This study was based on a pooled group of patients participating in two single-site randomized controlled trials (ClinicalTrials.gov identifier: NCT01812564 and NCT02104258). The first trial (the growth factor study) investigated the effect of platelet-rich plasma injections and has been completed, with 90 participants included between 2009 and 2013.(20) The second trial (the Aspetar hamstring rehabilitation study) was initiated in 2014, investigating the effect of two rehabilitation protocols with different emphasis on eccentric (lengthening) exercises on time to RTS and rate of re-injury. This trial is ongoing, and at the time of analysis (August 2018), 85 out of 90 participants had been randomized and completed their treatment.

Eligibility criteria

Data were collected between November 2009 and July 2018 from players meeting the eligibility criteria in the respective rehabilitation trials (table 1). For the purpose of the current study, professional soccer players competing in Qatar were eligible. Each player underwent an annual periodic health evaluation (commonly referred to as screening) at the Aspetar Orthopaedic and Sports Medicine Hospital in Doha, Qatar. As part of the musculoskeletal component of the periodic health evaluation, all players received a comprehensive musculoskeletal test battery aimed at identifying potential biomechanical and anatomic risk factors for lower extremity injuries at the rehabilitation department of Aspetar.(21) Players that had isokinetic testing performed during their annual periodic health evaluation and suffered a subsequent hamstring injury that was included in either of the two rehabilitation trials, were included in the current study. All players signed a consent form ahead of inclusion. Ethical approvals were obtained from the Institutional Review Board, Anti-doping Laboratory, Qatar (IRB F2013000003 and IRB F2013000009).

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Diagnosis of an acute hamstring injury was performed by a sports medicine physician and included a comprehensive patient history, physical examination, and confirmatory MRI. In both trials, athletes were included if a positive clinical examination was confirmed by a positive MRI diagnosis. (Note that the described inclusion criteria between the two trials differ slightly as presented in table 1.) Radiological grading of the injuries was performed by the same radiologist using the most commonly accepted grading system defining four grades: grade 0 with no pathological findings, grade 1 with muscle oedema only but without tissue damage, grade 2 as partial muscle tear and grade 3 with a complete muscle tear.(22) Players with negative MRI results, even if they had a positive history and physical signs of hamstring injury (e.g. acute mechanism of onset, palpable tenderness, and localized pain on resisted hamstring contraction), were excluded.

Isokinetic testing

Isokinetic strength testing (System 3, Biodex, New York, USA) was performed on three separate occasions: 1) during the annual periodic health evaluation (both limbs) pre-injury, 2) during the initial examination within seven days post-injury (contralateral uninjured limb only), and 3) prior to the final RTS assessment (both limbs).

Prior to isokinetic testing, the athletes were instructed as to the nature and purpose of the isokinetic testing. They performed a standardized warm-up procedure comprising five minutes on a stationary exercise bike (Technogym, Italy) at a resistance (in W) equivalent to 1.5 times their bodyweight (in kg) at their chosen cadence (typically this was approximately 85 rpm). Prior to each isokinetic test, the athlete was instructed as to the mode of testing, and given a minimum of three practice repetitions, and testing was not started unless the athlete thought they were ready to do so. For the periodic health evaluation, the order (i.e. left, right) was randomized, and

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this was maintained for all the different modes and speeds for that athlete. At discharge, the uninjured limb was tested first, followed by the injured limb. During testing, vigorous verbal encouragement was provided.

Testing comprised three modes and speeds. First, the players were tested over five repetitions of concentric knee flexion and extension at 60°/s, followed by 10 repetitions of concentric knee flexion and extension at 300°/s. These test modes measured the concentric strength of the "quadriceps" (knee extension) and "hamstring" (knee flexion) muscles. Finally, players performed five repetitions of eccentric knee extension at 60°/s which measured the eccentric strength of the knee flexors (principally the "hamstring" muscles). The reliability of the isokinetic tests has been described previously.(19)

Standardized rehabilitation protocols

The participants in both randomized control trials completed comparable standardized criteria-based rehabilitation protocols. The protocols included range of motion exercises, progressive strengthening exercises and an incremental, progressive return to running program. The rehabilitation was performed at one location, supervised by one of six sports physical therapists with 7–25 years of experience treating athletes. The (treating) physiotherapist was blinded to the MRI findings. In both trials functional, criteria-based progressions (as opposed to time-based progressions) were utilized for the six-stage rehabilitation protocols.(17,23) The rehabilitation protocol for the first study has been published,(17) and the structure of the two different protocols used in the second trial is freely available online: (https://www.youtube.com/watch?v=Fzex_zG1JtA; http://www.aspetar.com/AspetarFILEUPLOAD/UploadCenter/636209313253275549_Aspetar%20Hamstring%20Protocol.pdf)

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Soccer-specific training

After successfully finishing the first three stages of the rehabilitation protocol, the final stages of soccer-specific training were supervised by a sports rehabilitator with more than 10 years of experience in professional soccer. The protocol consisted of three stages (typically completed in three days) and progression of volume and intensity of direction changes, sprints, jumps, passes, shooting, interval running, one-on-one attacking and defending drills. The training was designed to mimic muscle fatigue and competitiveness during soccer training and game situations. Successful completion of the soccer-specific training was defined as full sports-specific functional ability without obvious limitations and/or symptoms.(17)

Return to sport

The player had to complete the soccer-specific training before the RTS assessment was performed. The treating sports medicine physician was informed of the physical therapists' RTS assessment findings, which included clinical examinations, as well as any comments from the sports rehabilitator regarding the on-field sessions. With this information (from the physical therapist and sports rehabilitator), the sports medicine physician then took a history, performed a clinical examination, and then decided on the athlete's readiness to RTS. RTS was defined as the number of days between initial injury and return to full unrestricted training and/or match play.(17) Acute hamstring strain injuries at the same site that occurred within 2 months from RTS, were classified as re-injuries.(4,24)

Statistical analyses

All statistical analyses were performed with Excel 2016 (Microsoft, Redmond, USA) and SPSS software (V.20.0; SPSS, Chicago, Illinois, USA). Continuous variables were expressed as mean with standard deviation (SD). We analyzed differences in binomial data between groups with an

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unpaired t test. Significance was assumed if the p value was less than 0.05. The percentage was calculated of players with 5%, 10%, 15%, 20% and 25% isokinetic strength difference when comparing 1) the contralateral limbs pre-injury, 2) the contralateral limbs at RTS, and 3) the ipsilateral limb pre-injury to RTS.

RESULTS

The inclusion of players into the study is outlined in figure 1. In total, 41 soccer players (age 25 ± 4 yrs, height 175 ± 6 cm, body mass 71 ± 7 kg, body mass index 23 ± 2 kg/m^2) were included. There was no significant difference in any of these measures between the 72 excluded players (age 24 ± 5 yrs, height 177 ± 6 cm, weight 73 ± 9 kg, body mass index [BMI] 23 ± 2 kg/m^2) and the 41 players included (p >0.05).

Of the 41 injuries, 21 were classified as grade I injuries and 20 as grade II, with 73.1% located in the biceps femoris long head, 22.0% in the semimembranosus, and 4.9% in the semitendinosus.

The injured players included six forwards, 17 midfielders, 17 defenders, and one goalkeeper. The players included in this analysis took on average 25.3 ± 8.9 days to RTS. One player suffered a re-injury from the completed trial.

The isokinetic strength of the players pre-injury during the periodic health evaluation (both limbs), post-injury at initial assessment (uninjured limb only), and at RTS (both limbs) is presented in figure 2, demonstrating only small changes in strength between the measurements. Quadriceps and hamstrings concentric strength at 60°/s demonstrated statistically significant differences of 4.9% and 5.3% respectively when comparing the injured limb at RTS to pre-injury (table 2).

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At RTS, the average strength of the injured limbs was >95% normalized to the pre-injury strength for all measurements (table 2). The side-to-side differences in isokinetic strength are presented as percentages for isokinetic strength pre-injury (panel A) and at RTS (panel B). The same cut-offs are used when comparing the change in strength at RTS to pre-injury in the injured limb (panel C); figure 3 demonstrating a negative difference (i.e. the injured leg was weaker) and figure 4 a positive difference (i.e. the injured leg was stronger). The greatest difference (non-significant with large CIs) was found for the cut-offs of >20% when comparing panel C to panels A and B (figure 3).

Overall, 63.4% of the players had a >10% negative side-to-side strength difference (i.e. injured limb being weaker) and 57.9% of the players had a >10% positive strength difference (i.e. injured limb being stronger) in at least one of the five isokinetic variables (figures 3 and 4).

DISCUSSION

In this series of 41 professional soccer players with an MRI-confirmed hamstring injury, novel analyses, comparing isokinetic strength of the injured limb at RTS to pre-injury strength measures, are presented, as well as between-limb comparisons. The present results suggest that a 10% difference (positive or negative) is present in a large proportion of players at RTS following hamstring injury, and the distribution of isokinetic strength variables in the injured limb at RTS are similar to pre-injury strength measures. Additionally, the side-to-side strength differences are similar at RTS when compared to pre-injury strength differences.

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Return to sport decision making after hamstring injury

Only one previous study has investigated isokinetic strength at RTS, suggesting that restoration of isokinetic strength (compared to the uninjured limb) does not seem to be required for the successful completion of a criteria based rehabilitation program.(17) However, this comparison was made between limbs, comparing the injured to the uninjured side. The present study is the first to compare RTS strength to pre-injury strength in the same limb.

Making RTS decisions following hamstring injuries is challenging. Different criteria have been suggested, but there is a lack of validated criteria and poor consensus on which measurements are useful to determine readiness to RTS.(9) Although evidence to support a specific recommendation is lacking, a cut-off of 10% compared to the uninjured side continues to be prevalent in research studies and clinical practice.(13) However, the present results suggest that a 10% difference (positive or negative) is present in a large proportion of players at RTS following hamstring injury and is even observed in a similar proportion of players pre-injury.

Regardless of statistical significance, the small changes in the isokinetic measures and large confidence intervals suggest that it would be clinically meaningless to impose a specific cut-off value as a criterion for RTS (figure 2). Furthermore, it is clear from figure 3C that a large percentage of players displayed strength changes in the injured limb at RTS compared to pre-injury strength for all the isokinetic variables. The side-to-side differences observed pre-injury and at RTS (figure 3A and 3B) are similar to the change in the injured limb (figure 3C), apart from a greater percentage of players demonstrating a >20% difference at RTS. It remains difficult to determine which isokinetic strength cut-offs are useful to determine when players are ready for RTS, or indeed, which players are at greater risk of re-injury.

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Risk of re-injury

Risk of re-injury is considered a potential decision modifier,(7) with recurrent injuries typically reported to occur early after RTS.(25–28) Isokinetic strength differences with specific cut-offs are thought to identify players at greater risk of re-injury, and therefore promoted as a criterion not only in RTS,(29) but also when used to determine progression during a criteria-based rehabilitation program.(12) As the second trial from which we included players is still ongoing, we were unable to investigate re-injury data, and only one of the players from the completed trial suffered a re-injury within the first two months.

However, a previous investigation in the same cohort as this present study (53.7% of the players in the current study were also included in the previous analysis) reported no differences in isokinetic strength at RTS between those players who experienced hamstring re-injury, and those who did not.(17) A recent randomized trial among recreational athletes, comparing early and delayed rehabilitation, also found no difference in isokinetic strength at RTS, reporting just one re-injury for the entire cohort.(30) These findings(17,30) suggest that isokinetic strength may not need to be fully restored to limit re-injury risk if a criteria-based rehabilitation program is completed successfully.

Isokinetic strength differences

Isokinetic side-to-side strength differences at RTS are commonly presented as deficits.(11,17,29) This may reflect a bias in our view of strength profiles, as strength is preferentially reported as what is “missing”, i.e. the injured leg being weaker. However, as demonstrated in figure 4, a similarly large proportion of players present with a positive strength difference, i.e. the injured leg being stronger. Thus, approximately 60% of the players had at least one negative or positive

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isokinetic hamstring strength difference of more than 10%. Strength differences, whether positive or negative, seem to be a standard finding for isokinetic dynamometry testing, and so-called "deficits" should be interpreted with caution.

There were no significant differences in strength of the uninjured limb for any of the isokinetic modes of testing pre-injury compared to the post-injury tests during the initial assessment (figure 2). Although we observe a slight decrease in strength in the uninjured limb post-injury, the reason for this is not known. Quadriceps and hamstrings concentric strength at 60°/s demonstrated a statistically significant change in the injured limb (table 2) between RTS and pre-injury strength, but the small absolute differences are well within the minimal detectable change values for these modes of isokinetic testing.(29) Together with the large confidence intervals, these small differences indicate little clinical value in this finding.

Variability

The asymmetries presented may form useful reference data for clinicians. Importantly, these data should be interpreted in light of the variability within isokinetic testing that has been previously documented.(29) In figure 3 and 4, where we compare the strength at RTS to pre-injury, as well as side-to-side differences, the 95% CI includes 50% of the population for any of the varying cut-points from 5% to 20%.

If we consider that the minimal detectable change in peak torque for all these measures is greater than 20%,(19) it is apparent that the previously suggested 10% cut-point is likely overly conservative; beyond the capability of the measurement due to a large amount of intra-individual variability. Recently, we have also shown that isokinetic strength varies by 15% from one pre-season to the next.(31) If we were to compare the pre-injury and RTS strength, these results

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would need to be interpreted in light of the substantial variability we might expect within individual players over time.

It seems that 20% is more likely a reasonable 'cut-point' for when an isokinetic strength difference represents a true difference, however, this cut-off requires further investigation as a potential risk factor. The use of isokinetic testing in isolation is not supported, and can at best only be considered a component in RTS decision making.(32)

Strengths and limitations

A major strength of this study is that all tests performed utilized the same isokinetic testing system with highly experienced assessors, and it was performed in a single clinical setting for professional athletes. All players underwent standardized criteria-based rehabilitation protocols, together with a standardized and comprehensive RTS assessment. The hamstring injuries included were confirmed with both clinical examinations and positive MRI results. However, since only grade I and II injuries were included, it remains unknown if our results can be extrapolated to MRI negative or grade III (complete avulsions) hamstring injuries.

The isokinetic dynamometry strength testing was performed at the end of the final soccer specific training session as part of the RTS assessment. Therefore, players may have been somewhat fatigued at the time of testing, and it is not known whether this may have influenced the results. Although the isokinetic test results were not used as a set criterion in the RTS decision, the sports medicine physician were not blinded to the results. Since we could not control the weighting and influence of every bit of information used in the process, we cannot be sure how the isokinetic results, or indeed a number of other factors, may have influenced the RTS decision.

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An important limitation in this study is incomplete re-injury data. Although only one re-injury was registered from the completed trial, we do not know the re-injury rate in the second trial.

This study was performed in a multinational, multi-language setting, and while every effort was made to guarantee that players comprehended the test procedure and directions, it is possible that some players did not fully understand the instructions. We also acknowledge the homogeneity of our study population of professional male soccer players, which limits the generalization of these findings to other sports, age groups or female players.

Clinical implications

The use of isokinetic strength testing in isolation as a criterion for RTS is not supported by these data. Importantly, side-to-side isokinetic strength differences are common at RTS, and when compared to pre-injury strength. The injured limb presents with similar strength differences when comparing strength at RTS to pre-injury measures, and in a large proportion of players it might favor the injured limb. The previous accepted cut-off of 10% is inappropriate, as side-to-side strength differences of 5-20% are commonly found, both at RTS and pre-injury.

CONCLUSION

The distribution of positive and negative isokinetic strength differences when comparing strength at RTS to pre-injury measures were similar, urging clinical caution when interpreting these results. Small changes in isokinetic strength of the injured limb at RTS compared to pre-injury strength suggest that isokinetic cut-off values are not useful to guide restoration of strength as a criterion for RTS after hamstring injury.

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List of co-authors and their contribution:

Prof Erik Witvrouw – Concept and outline as well as writing and editing of the manuscript

Prof Roald Bahr – Concept and outline as well as writing and editing of the manuscript

Dr. Johannes Tol -Concept and outline as well as writing and editing of the manuscript

Dr. Robin Vermeulen - Investigator, statistical and data analyses, writing and editing of the manuscript

Dr. Rod Whiteley – Investigator, data analyses, data collection and writing and editing of the manuscript

Dr. Arnlaug Wangensteen - Investigator, concept and outline, data collection, and writing and editing of manuscript

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Figure 3 Proportion of players (y-axis) that demonstrate a negative isokinetic side-to-side strength difference (injured leg weaker) for the five different peak torque measurements (x-axis) with varying cut-points from 5% up to 25%. Data are presented for pre-injury (panel A), return to sport (panel B), as well as pre-injury strength compared to same-side return to sport strength (panel C) as percentages with 95% CI. HConc, hamstrings concentric; HEcc, hamstrings eccentric; QConc, quadriceps concentric.

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Table 1 Eligibility criteria for the two randomized control trials.

Table 2 The isokinetic strength of the injured players' limbs pre-injury compared to return to sport. Absolute values and mean force values are shown in Nm with standard deviation (SD) or percentages.

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Figure 1

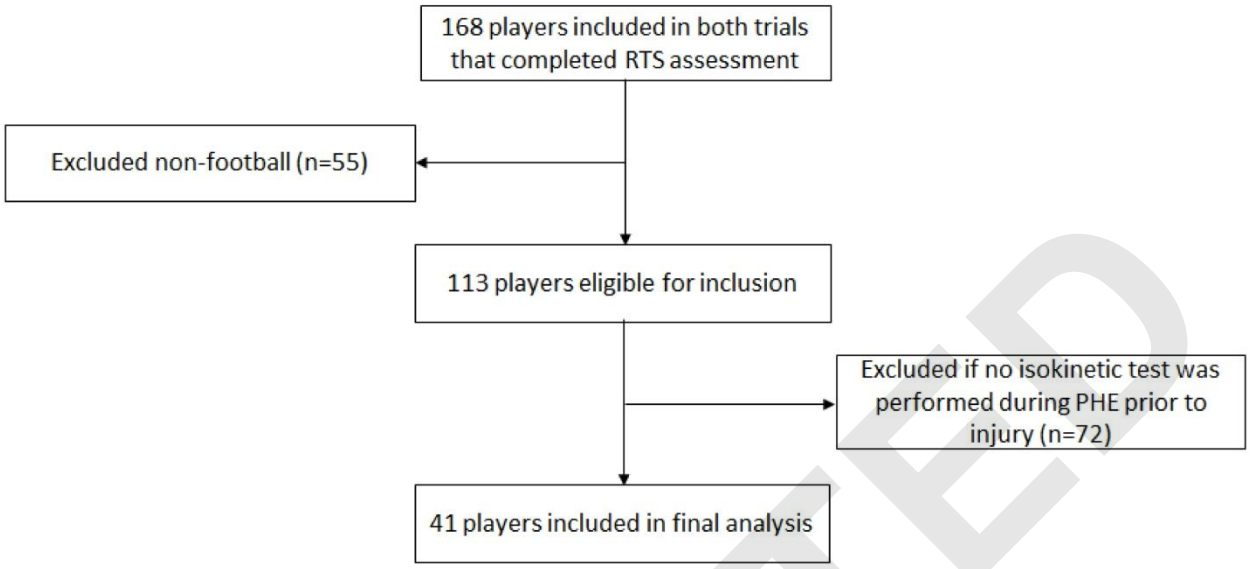

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Figure 2

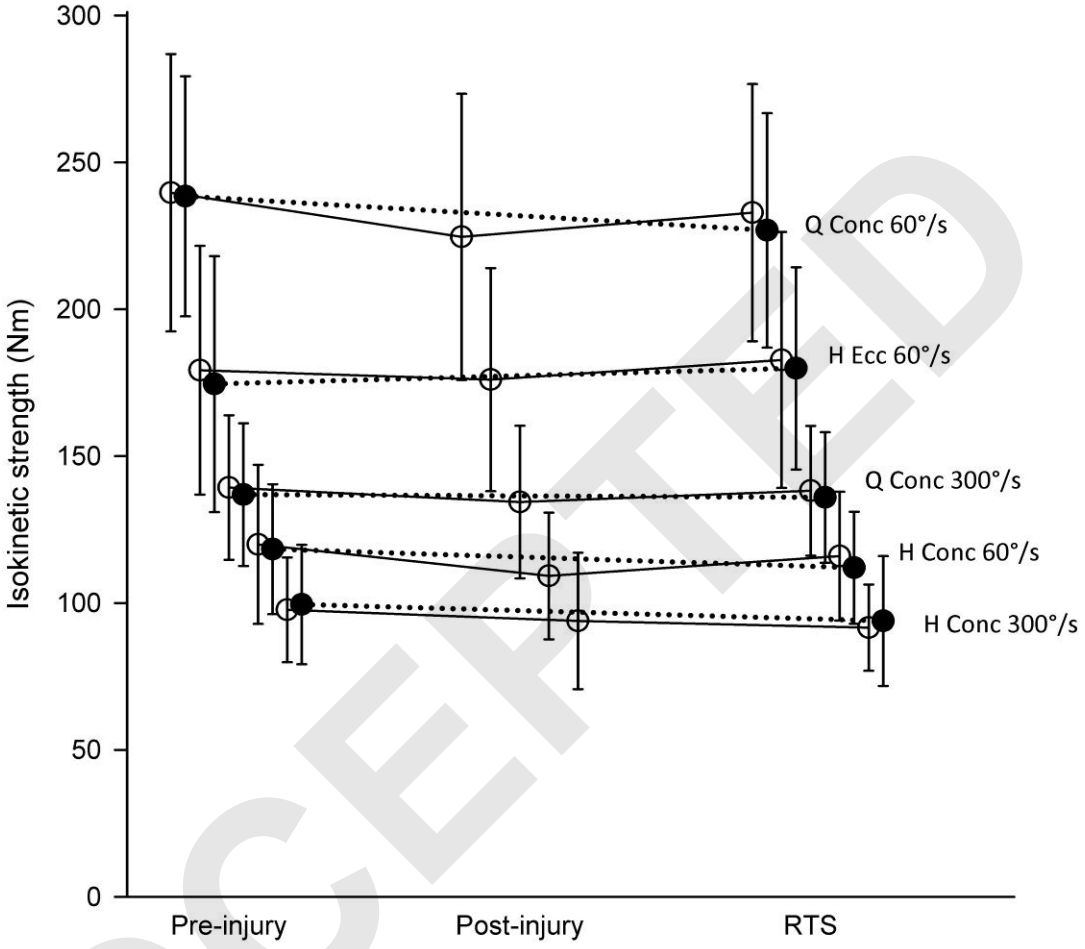

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Figure 3

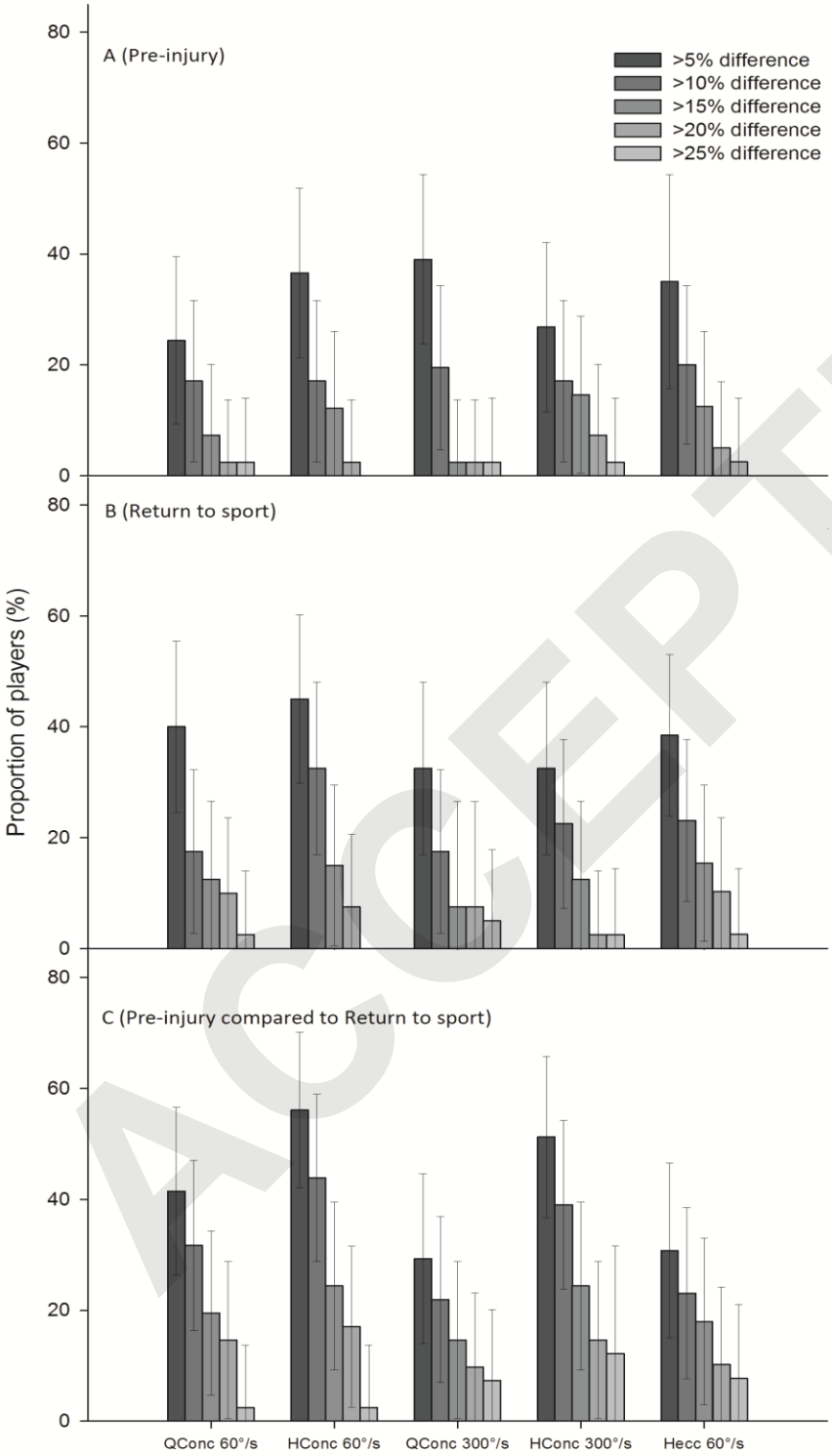

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Figure 4

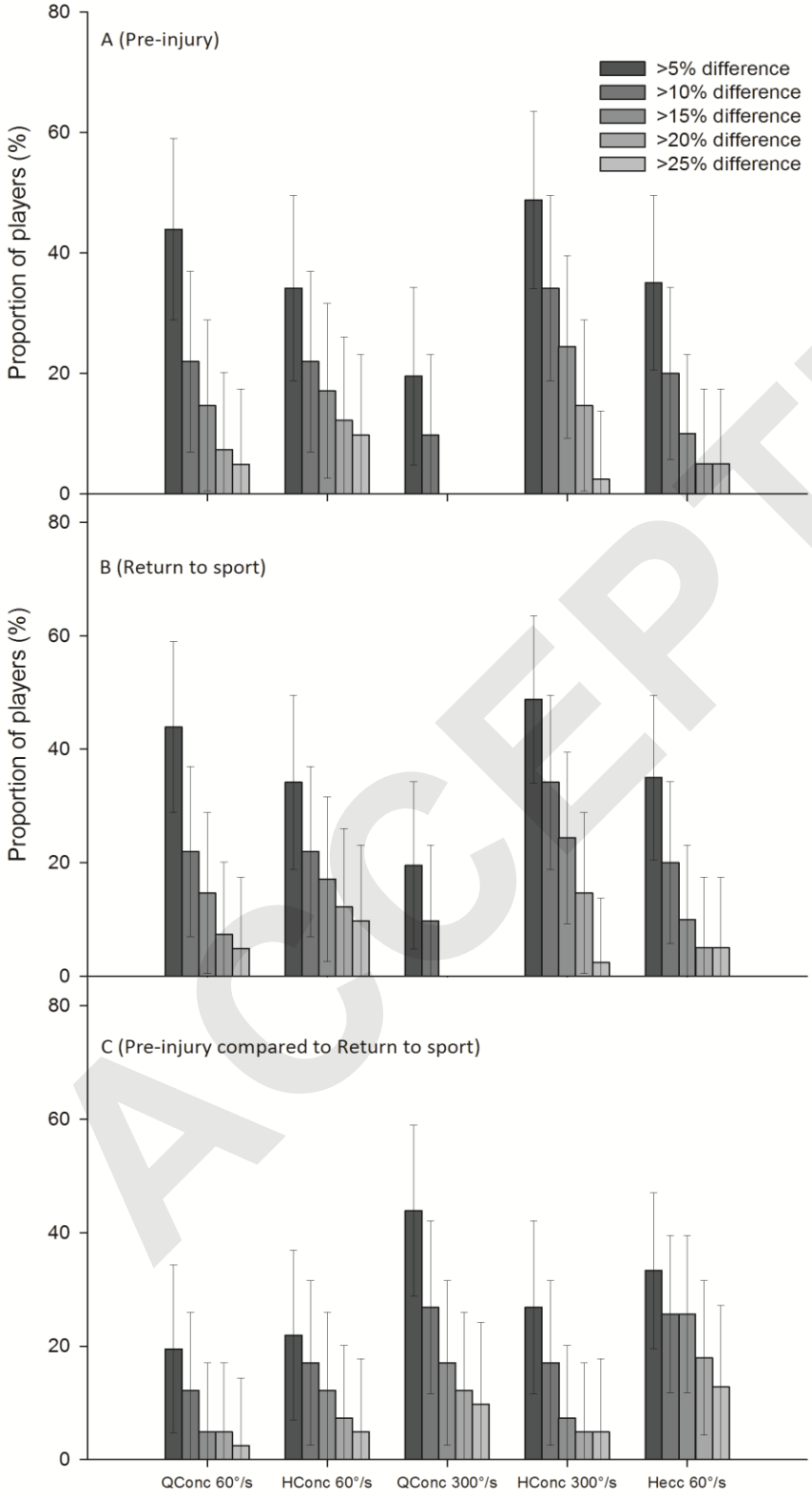

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Table 1 Eligibility criteria for the two randomized control trials.

Growth factor study	Aspetar hamstring rehabilitation study
Inclusion criteria ▪ Male gender ▪ Age 18-50 years ▪ Acute onset of posterior thigh pain ▪ Presenting and MRI within five days from injury ▪ MRI confirmed grade I or II hamstring lesion ▪ Able to perform five sessions of physiotherapy a week at our clinic ▪ Available for follow-up *Exclusion criteria* ▪ Contraindication to MRI ▪ Re-injury ≤2 months after RTS or chronic hamstring injury >2 months ▪ Other concurrent injury inhibiting rehabilitation ▪ Unwilling to comply with follow-up ▪ Needle phobia ▪ Overlying skin infection	*Inclusion criteria* ▪ Male athletes ▪ Age 18-50 years ▪ Acute onset posterior thigh pain when training or competing, identified as: - patient reported incident event - patient reported pain in posterior thigh ▪ Clinical diagnosis of an acute hamstring muscle strain injury, defined as: - localised pain during palpation of hamstring muscle - increasing pain during isometric contraction - localised pain when performing a passive straight leg test ▪ MRI confirmed isolated hamstring lesion ▪ MRI performed ≤5 days from injury ▪ Available for ≥3 physiotherapy sessions per week at Aspetar ▪ Available for follow-up

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▪ Diabetes, immune-compromised state ▪ Medication with increasing bleeding risk ▪ Medical contraindication to injection	*Exclusion criteria* ▪ Re-injury ≤2 months after RTS ▪ Chronic hamstring complaints >2 months ▪ Grade III hamstring tear ▪ Contraindications to MRI ▪ Already included with prior injury

RTS, return to sport, MRI, magnetic resonance imaging.

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Table 2 The isokinetic strength of the injured players' limbs pre-injury compared to return to sport. Absolute values and mean force values are shown in Nm with standard deviation (SD) or percentages.

	Pre-injury (Nm)	**RTS (Nm)**	**Difference (Nm)**	**RTS compared to Pre-injury (%)**	**P value**
INJURED LIMB					
Quadriceps					
Concentric @60°/s	238.5 (40.9)	226.9 (39.9)	11.6	95.8 (64.3 to 126.7)	**0.03**
Concentric @300°/s	136.9 (24.3)	135.9 (22.2)	1.0	99.8 (61.2 to 134.2)	0.78
Hamstrings					
Concentric @60°/s	118.3 (22.1)	112.0 (19.1)	6.3	95.9 (68.3 to 152.5)	**0.03**
Concentric @300°/s	99.5 (20.3)	93.9 (22.1)	5.6	95.2 (60.3 to 127.8)	0.14
Eccentric @60°/s	174.5 (51.3)	175.5 (44.1)	-1.0	100.1 (55.3 to 155.8)	0.91
UNINJURED LIMB					
Quadriceps					
Concentric	239.7 (47.2)	232.9	6.8	101.3 (61.9 to 160.4)	0.53

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@60°/s		(43.8)			
Concentric @300°/s	139.3 (24.6)	138.2 (22.1)	1.1	102.9 (65.9 to 186.6)	0.94
Hamstrings					
Concentric @60°/s	120.0 (27.1)	116.0 (21.9)	4.0	102.7 (51.2 to 147.7)	0.53
Concentric @300°/s	97.7 (17.8)	91.6 (14.7)	6.0	95.5 (56.9 to 147.5)	0.09
Eccentric @60°/s	179.3 (42.4)	182.8 (43.6)	-3.5	100.4 (59.1 to 181.6)	0.77

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